

RL-Guided Multi-Objective Evolution

PPO - Evolutionary Action Guidance

Actor
Selects Evolutionary
Action

← Advantage →

Critic
Estimates Action Value

Evolutionary Action

Evolutionary Search

Population

Mutation + Crossover
Grammar-constrained actions

Symmetry + Validity Check

Candidate Morphology

MuJoCo Physics Evaluation

3D folding

Magnetic actuation
B-Sweep Gait

Pheromone Stimuli
Response
Attracted/Repelled

**Multi-Objective
Fitness**

NSGA-III
Non-Dominated Sorting
Reference-Direction Niching

Next Generation

Next iteration

Reward

